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Communications network assignment

Question1

1.1)

Position 1: c1

Position 2: c2

Position 4: c4

Position 8: c8

Position	1	2	3	4	5	6	7	8	9	10	11	12
Bit	C1	C2	d1	C4	D2	D3	D4	C8	D5	D6	D7	D8

Data bits from 0010 0100:

d1 = 0, d2 = 0, d3 = 1, d4 = 0, d5 = 0, d6 = 1, d7 = 0, d8 = 0

c1 (odd positions: 1, 3, 5, 7, 9, 11):

Positions: 1, 3, 5, 7, 9, 11

Bits: c1, d1, d2, d4, d5, d7

Values: c1, 0, 0, 0, 0, 0

c1 = 0 0 0 0 0 = 0

c2 (positions with bit 2 set: 2, 3, 6, 7, 10, 11):

Positions: 2, 3, 6, 7, 10, 11

Bits: c2, d1, d3, d4, d6, d7

Values: c2, 0, 1, 0, 1, 0

c2 = 0 1 0 1 0 = 0

c4 (checks positions with bit 3 set: 4, 5, 6, 7, 12):

Positions: 4, 5, 6, 7, 12

Bits: c4, d2, d3, d4, d8

Values: c4, 0, 1, 0, 0

c4 = 0 1 0 0 = 1

c8 (checks positions with bit 4 set: 8, 9, 10, 11, 12):

Positions: 8, 9, 10, 11, 12

Bits: c8, d5, d6, d7, d8

Values: c8, 0, 1, 0, 0

c8 = 0 1 0 0 = 1

so c1 = 0 , c2 = 0 , c4 = 1 , c8 = 1

1.2)Position: 1 2 3 4 5 6 7 8 9 10 11 12

Bit: c1 c2 d1 c4 d2 d3 d4 c8 d5 d6 d7 d8

Value: 0 0 0 1 0 1 0 1 0 1 0 0

Question 2

2.1)

First octet = 132

falls under class B

2.2)

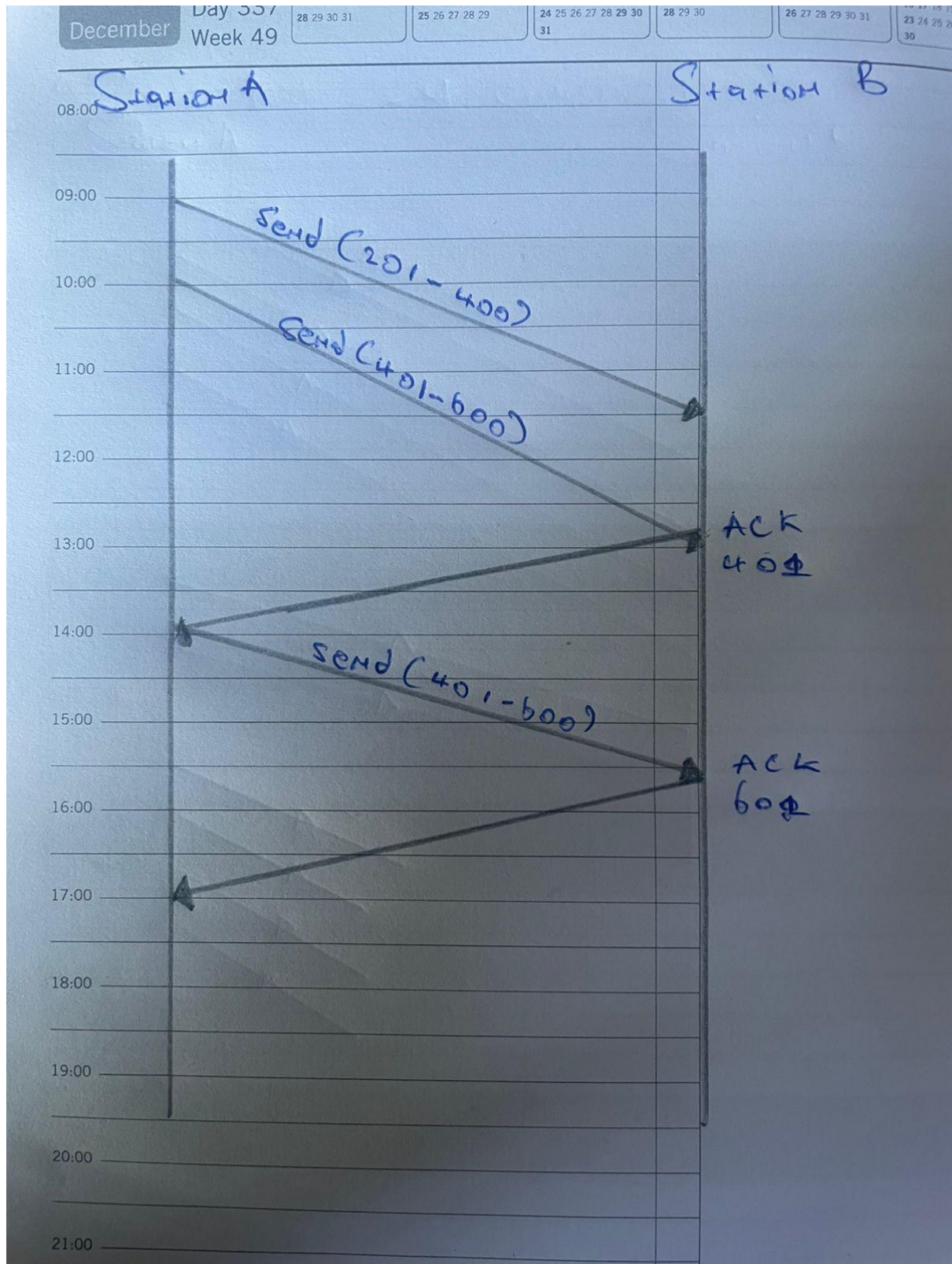
32 bits – 25 = 7

2 to the power of 7 = 128 addresses

2.3) 11111111.11111111.11111111.10000000

2.4) 32 - 25 = 7 bits

Question 3



Question 4

4.1 Impulse noise – Install special filters for analog signals implement digital signal processing for digital signals.

4.2 Crosstalk – Install proper shielding on cables

4.3 Echo – Install proper termination of cables

4.4 Jitter – User better quality electronic circuitry, use fewer repeaters, slow the transmission speed

Question 5

Toss the frame/packet - If a data frame arrives at a frame

relay switch and an error is detected after the cyclic checksum is performed, the frame is simply discarded. The rationale behind this action is twofold.

Return a message- sending a message back to the transmitter-is probably the most common form of error control. Returning a message was also one of the first error control techniques developed and is closely associated with a particular flow control technique.

Correct the error- use a process called forward error

correction--redundant information must be present so that the receiver knows which bit or bits are in error and what their original values were.

Question 6

6.1)An IEEE protocol (802.3ad-2000) which typically runs in most LAN devices can support link aggregation

Link aggregation attempts to balance the flow of messages over the multiple paths, unless the flow of frames belong to a particular conversation; then the frames are sent over one of the links.

6.2) - Increased bandwidth by combining links

-Enhances scalability and simplifies management

- Provides fault tolerance
- Support load balancing of traffic across links

Question 7

7.1 Error detection – finding whether an error occurred during transmission

Error control – overall process that includes detection, notification and correction

Error correction – fixing the error, either by the receiver or by requesting retransmission

7.2 Station – a device in a wireless network

Node – any device on a network that can send/receive data

7.3 Hop count- the number of routers a packet has already passed through

Hop limit – the maximum number of hops a packet is allowed before being discarded

7.4DHCP – automatically assigns IP addresses to devices on a network

NAT – translates private IP addresses to a public IP for internet

7.5 Shared segment network - Multiple devices share the same bandwidth

Dedicated segment network – each device has a dedicated bandwidth path

Question 8

8.1 IEEE 802.3 CSMA/CD

8.2 64 bytes

8.3 Destination Address: Specifies the MAC address of the receiving device.

Source Address: Specifies the MAC address of the sending device.

Therefore : These fields are used to identify the sender and intended receiver on a LAN

8.4 Each field is 6 bytes long because a MAC address is 48 bits (6 bytes) in length.

8.5 is used for error detection.

It allows the receiver to detect if the frame was corrupted during transmission using a Cyclic Redundancy Check

Question 9

9.1 connection – orientated communication

Fixed route during session

Reliable delivery

9.2 Reliable delivery – Packets arrive in sequence, reducing the need for reordering.

Lower overhead – Shorter headers are used since routing info is established during setup.

9.3 Connection setup delay – Time is needed to establish the virtual circuit before data can be sent.